

INVERSE ROMAN DOMINATION IN GRAPHS

M. KAMAL KUMAR

*VTU Belgaum, Karnataka, India
 kamalmvz@rediffmail.com*

L. SUDERSHAN REDDY

*Department of Mathematics
 CMR Institute of Technology, Bangalore, India*

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Motivated by the article in Scientific American [7], Michael A Henning and Stephen T Hedetniemi explored the strategy of defending the Roman Empire. Cockayne defined Roman dominating function (RDF) on a Graph $G = (V, E)$ to be a function $f : V \rightarrow \{0, 1, 2\}$ satisfying the condition that every vertex u for which $f(u) = 0$ is adjacent to at least one vertex v for which $f(v) = 2$. For a real valued function $f : V \rightarrow R$ the weight of f is $w(f) = \sum_{v \in V} f(v)$. The Roman domination number (RDN) denoted by $\gamma_R(G)$ is the minimum weight among all RDF in G . If $V - D$ contains a roman dominating function $f^1 : V \rightarrow \{0, 1, 2\}$. “D” is the set of vertices v for which $f(v) > 0$. Then f^1 is called Inverse Roman Dominating function (IRDF) on a graph G w.r.t. f . The inverse roman domination number (IRDN) denoted by $\gamma_R^1(G)$ is the minimum weight among all IRDF in G . In this paper we find few results of IRDN.

Keywords: Domination number; inverse domination number; roman domination number.

Mathematics Subject Classification: 05C69

1. Introduction

The theory of Domination was introduced by Claude Berge in 1958. The inspiration for this concept was drawn from the classical problem of covering chessboard with minimum number of chess pieces.

Motivated by the article written by Ian Steward titled “Defend the Roman Empire” [7], Henning and Hedetniemi explored the strategy of defending the Roman Empire. They believe that the Roman Empire had the potential of saving the Emperor, Constantine the Great, curtailing costs of maintaining legions while still defending the Roman Empire [9].

Mean while Cockayne defined roman dominating function (RDF) on a graph $G = (V, E)$ to be a function $f : V \rightarrow \{0, 1, 2\}$ satisfying the condition that every

vertex u for which $f(u) = 0$ is adjacent to at least one vertex v for which $f(v) = 2$. For a real valued function $f : V \rightarrow R$ the weight of f is $w(f) = \sum_{v \in V} f(v)$. The roman domination number (RDN) denoted by $\gamma_R(G)$ is the minimum weight among all RDF in G . In another words, a roman dominating function is a coloring of the vertices of a graph with the colors $\{0, 1, 2\}$ such that every vertex colored “0” is adjacent to at least one vertex colored “2”.

Inverse roman dominating function is also a roman dominating function, if set $V - D$ contains a roman dominating function $f^1 : V \rightarrow \{0, 1, 2\}$. D is the set of vertices v for which $f(v) > 0$. Then f^1 is called Inverse Roman Dominating function (IRDF) on a graph G w.r.t. f . The inverse roman domination number (IRDN) denoted by $\gamma_R^1(G)$ is the minimum weight among all IRDF in G . For any undefined terms or notation in this paper, we refer Harary [6].

2. Preliminary Results

Theorem 2.1. For any given graph G , $\gamma_R(G) \leq \gamma_R^1(G)$.

Proof. An inverse roman domination is also a roman domination hence $\gamma_R(G) \leq \gamma_R^1(G)$ i.e., while the vertices are labeled with positive weight for finding roman dominating function vertices having max degree is chosen. Hence in $|V - D|$ the vertices left with degree less than or equal to D is chosen for finding inverse roman dominating function. $\square$

Theorem 2.2. For any complete graph G with n vertices, $\gamma_R(K_n) = \gamma_R^1(K_n) = 2$.

Proof. In a complete graph for finding roman dominating function one vertex is labeled 2 and all other vertices are labeled 0 as the vertices are adjacent to each other therefore $|D| = 1$ in $|V - D|$ there exist a vertex whose degree is $n - 1$ hence $\gamma_R^1(K_n) = 2$. $\square$

Theorem 2.3. For the classes of cycle C_n with n vertices, $\gamma_R(G) = \gamma_R^1(G) = \lceil \frac{2n}{3} \rceil$.

Proof. Label the vertices of cycle C_n as $u_1, u_2, u_3, \dots, u_n$. Let D be a roman dominating set. In every minimal RDF there will be no two adjacent vertices with positive weights. Hence inverse roman dominating function can be obtained by rotating one vertex clockwise of the vertices of roman dominating function. Then D^1 is the inverse roman domination set, $D^1 = \{u_i/u_{i+1} \in D\}$ hence $|D^1| = |D|$, hence $\gamma_R(G) = \gamma_R^1(G) = \lceil \frac{2n}{3} \rceil$. $\square$

Theorem 2.4. For the classes of paths P_n with n vertices,

$$\gamma_R^1(G) = \begin{cases} \left\lceil \frac{2n}{3} \right\rceil + 1 & \text{if } n \equiv 0 \pmod{3} \\ \left\lceil \frac{2n}{3} \right\rceil & \text{otherwise} \end{cases}.$$

Proof. Let P_n be a path with n vertices. Label the vertices of path P_n as $u_1, u_2, u_3, \dots, u_n$. $\square$

Consider two cases.

Case 1. If $n \equiv 0 \pmod{3}$ there exists a unique roman domination set $D = \{u_{3p-1}/1 \leq p \leq n/3\}$ with $f(u_i) = 2$ where $D^1 = \{u_{i+1}/u_i \in D\} \cup \{u_1\}$ with $f^1(u_i) = 2/\forall i \neq 1$ and $f^1(u_1) = 1$ therefore $\gamma_R^1(G) = |D^1| = |D| + 1$.

Case 2. If $n \not\equiv 0 \pmod{3}$ there exists a minimal roman domination set D in which end vertex u_n is included. Find inverse roman dominating function by reverse labeling i.e., $u_1, u_2, u_3, \dots, u_n$ are labeled in roman dominating function then $u_n, u_{n-1}, u_{n-2}, \dots, u_1$ are labeled for inverse roman dominating function. Therefore $D^1 = \{u_{n-i+1}/u_i \in D\}$. Hence $\gamma_R^1(G) = |D^1| = |D| = \gamma_R(G)$.

Theorem 2.5. For the classes of wheel W_n with $n \geq 4$ vertices, $\gamma_R^1(G) = \lceil \frac{2(n-1)}{3} \rceil$.

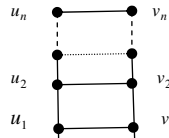
Proof. Label the maximum degree vertex as v_1 and remaining vertices as $v_2, v_3, v_4, \dots, v_n$. We know that $W_n = K_1 + C_{n-1}$, hence $\gamma_R(G) = 2$ and $|D| = 1$. Inverse roman domination number $\gamma_R^1(G)$ of W_n is the roman domination number of C_{n-1} . Therefore $\gamma_R^1(W_n) = \gamma_R(C_{n-1}) = \lceil \frac{2(n-1)}{3} \rceil$. $\square$

Theorem 2.6. For the classes of star $K_{1,n-1}$ with n vertices, $\gamma_R^1(G) = n$.

Proof. Label the maximum degree vertex as v_1 and remaining pendant vertices as $v_2, v_3, v_4, \dots, v_n$. We know that $\gamma_R(G) = 2$ i.e., $f(v_1) = 2$ and $f(v_i) = 0$ for $i \neq 1$, hence $|D| = 1$. For inverse roman dominating function of $K_{1,n-1}$ all the pendant vertices will be labeled 1 except one pendant vertices which will be labeled 2. Therefore $\gamma_R^1(K_{1,n-1}) = 1(n-2) + 2 = n$. $\square$

Theorem 2.7. The inverse roman domination number of a ladder $G_{2,n}$ is $\gamma_R(G) = \gamma_R^1(G) = n + 1$.

Proof. Label the ladder $G_{2,n}$ from the bottom to top as $u_1, u_2, u_3, u_4, \dots, u_n$ shown in the figure.



It can be shown that there is always a minimum roman dominating function where both of the vertices on a rung of a ladder would never both have a positive weight. Hence $\gamma_R(G) = \gamma_R^1(G) = n + 1$ by symmetry. i.e. $D = \{u_i \vee v_j/i, j = 1, 2, \dots, n\}$, then $D^1 = \{v_i \vee u_j \in D/i = 1, 2, \dots, n\}$. Hence $\gamma_R^1(G) = |D^1| = |D| = \gamma_R(G) = n + 1$. $\square$

Theorem 2.8. For any graph G of order n with maximum degree Δ and minimum degree δ , $\lceil \frac{2n}{\Delta+1} \rceil \leq \gamma_R^1(G) \leq n - \delta + 1$.

Proof. From Theorem 2.1, for any given graph G , $\gamma_R(G) \leq \gamma_R^1(G) \cdot \lceil \frac{2n}{\Delta+1} \rceil \leq \gamma_R(G)$ [14], hence the lower bound. While the vertices are labeled with positive weight for finding roman dominating function vertices having maximum degree is chosen. Hence in $|V - D|$ the vertices left with minimum degree is chosen for finding inverse roman dominating function. Without loss of generality we assume that the vertices in D^1 will be of minimum degree i.e., we have for $\gamma_R^1(G)$, $f^1(u) = 2$, $f^1(N(u)) = 0$ and $f^1(V(G) - N(u)) = 1$, $\deg(u) = \delta$. Therefore $\gamma_R^1(G) = 0(\delta) + 1(n - \delta - 1) + 2(1) = n - \delta - 1 + 2 = n - \delta + 1$. $\square$

3. Open Problem

- (1) Inverse Roman Domination can be explored for other classes of graphs.
- (2) Inverse Roman Domination can be connected to other types of domination parameters.
- (3) Results of Roman Domination can be compared and extended to Inverse Roman Domination.

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